

MELEUZ, Russian Federation

WMO: 289250

Lat: 52.9302N

Lon: 55.9174E

Elev: 592

StdP: 14.38

Time Zone: 5.00 (E05)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-19.9	-30.0	1.0	-25.1	-24.6	1.4	-19.7	26.4	20.9	21.9	21.9	1.0	200	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	89.4	68.9	85.9	67.5	82.7	66.3	72.6	84.5	70.1	81.5	67.9	79.3	6.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	106.0	79.8	66.0	98.0	77.4	63.8	90.5	74.0	36.6	84.7	34.4	81.8	32.5	79.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	17.1	14.9	-35.7	94.9	7.4	3.9	-41.0	97.7	-45.3	100.0	-49.4	102.2	-54.7	105.0
			-34.1	74.6	8.7	3.8	-40.4	77.4	-45.4	79.6	-50.3	81.7	-56.7	84.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	10.0	10.7	21.1	41.4	56.0	65.1	68.6	64.3	54.3	41.5	23.1	11.5	(6)
(7)		DBStd	24.25	13.58	14.18	10.61	11.60	8.58	7.42	5.98	7.08	8.49	8.85	12.16	15.01	(7)
(8)		HDD50	5923	1240	1099	895	305	43	1	0	0	49	292	807	1192	(8)
(9)		HDD65	9783	1705	1519	1360	708	295	91	32	98	332	729	1257	1657	(9)
(10)		CDD50	1961	0	0	0	48	230	454	578	443	179	29	0	0	(10)
(11)		CDD65	344	0	0	0	1	17	93	145	76	12	0	0	0	(11)
(12)		CDH74	3886	0	0	0	25	281	1060	1479	854	187	0	0	0	(12)
(13)		CDH80	1314	0	0	0	1	50	381	505	317	60	0	0	0	(13)
(14)	Wind	WSAvg	5.4	5.7	6.2	5.4	5.1	6.0	5.2	4.6	4.4	4.8	5.8	5.6	5.9	(14)
(15)	Precipitation	PrecAvg	17.9	1.5	1.1	1.4	1.0	1.3	2.2	1.7	1.6	1.2	1.9	1.5	1.7	(15)
(16)		PrecMax	23.1	2.4	2.3	3.4	2.4	3.9	4.0	3.1	3.1	1.9	3.9	3.1	4.2	(16)
(17)		PrecMin	11.6	0.4	0.4	0.2	0.0	0.1	1.0	0.1	0.2	0.4	0.3	0.2	0.5	(17)
(18)		PrecStd	3.2	0.7	0.6	1.0	0.7	0.9	1.0	0.9	0.7	0.6	1.0	0.9	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	36.7	44.8	77.4	85.2	94.9	92.9	93.5	87.7	70.7	47.9	35.1	(19)
(20)		MCWB	32.6	33.2	38.8	59.7	61.3	69.3	70.8	71.3	63.3	57.0	46.1	33.3	(20)	
(21)		2%	DB	32.2	33.4	39.1	71.8	81.7	88.2	88.6	88.0	78.5	65.0	42.4	33.5	(21)
(22)		MCWB	31.0	31.5	34.9	55.3	61.1	67.8	69.6	68.8	59.2	51.7	40.7	32.2	(22)	
(23)		5%	DB	30.2	31.9	36.9	66.5	77.6	84.1	85.6	83.0	73.9	59.9	39.5	32.4	(23)
(24)		MCWB	29.1	30.8	34.2	52.6	59.4	67.0	68.8	66.5	58.6	49.6	37.5	31.2	(24)	
(25)		10%	DB	27.9	29.8	35.0	60.2	72.7	80.4	82.5	78.2	69.4	55.4	37.0	30.7	(25)
(26)		MCWB	26.6	28.4	32.8	49.5	57.5	65.8	67.8	64.5	56.6	48.3	34.9	29.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	33.7	39.5	61.3	67.1	73.8	75.7	74.3	65.1	59.6	45.9	34.0	(27)
(28)		MDCB	33.9	35.5	43.2	76.3	78.6	86.6	86.9	88.8	82.3	67.7	47.6	34.8	(28)	
(29)		2%	WB	31.2	32.1	36.1	56.3	62.6	71.4	73.0	71.1	61.8	54.5	40.6	32.7	(29)
(30)		MDCB	32.1	33.1	38.3	69.1	77.1	82.7	84.6	83.0	75.5	61.3	41.8	33.5	(30)	
(31)		5%	WB	29.3	30.7	34.5	52.8	60.8	69.0	70.5	67.9	59.4	51.3	38.1	31.5	(31)
(32)		MDCB	30.3	32.0	36.5	64.5	74.7	80.5	81.8	78.7	70.8	57.6	39.4	32.3	(32)	
(33)		10%	WB	26.6	28.3	33.0	49.8	58.5	66.6	68.3	65.7	57.4	48.4	35.3	29.6	(33)
(34)		MDCB	27.8	29.7	35.0	59.4	70.5	77.6	79.8	76.7	67.9	55.0	36.5	30.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	13.6	15.4	15.6	18.5	20.9	20.8	19.7	20.7	19.7	14.6	10.7	13.4	(35)	
(36)		5% DB	MCDBR	9.3	10.7	12.3	27.9	29.0	27.0	25.4	28.0	28.2	24.4	10.6	10.6	(36)
(37)		MCWBR	8.8	9.9	9.9	14.3	13.0	12.2	10.7	12.0	13.4	14.2	9.2	10.4	(37)	
(38)		5% WB	MCDBR	9.3	10.2	10.4	26.3	25.8	24.1	23.4	24.9	25.8	22.0	8.1	9.9	(38)
(39)		MCWBR	9.1	9.7	8.8	14.5	13.3	13.1	12.0	12.6	13.6	14.3	7.9	10.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.326	0.379	0.427	0.407	0.399	0.411	0.432	0.381	0.353	0.326	0.298	(40)	
(41)		taud	2.050	2.009	1.931	2.045	2.245	2.327	2.294	2.215	2.324	2.303	2.173	2.033	(41)	
(42)		Ebn at Noon	202	236	247	253	267	271	264	250	250	230	193	175	(42)	
(43)		Edh at Noon	24	35	46	47	41	38	39	40	31	26	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	331	675	1122	1438	1758	1938	1885	1531	1033	512	235	213	(44)	
(45)		RadStd	39	58	115	142	152	208	159	141	126	100	49	41	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46) N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47) N/A

Nomenclature: See separate page